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Research Article

Engineering Silk Protein to Modulate Polymorphic Transitions for Green Lithography Resists

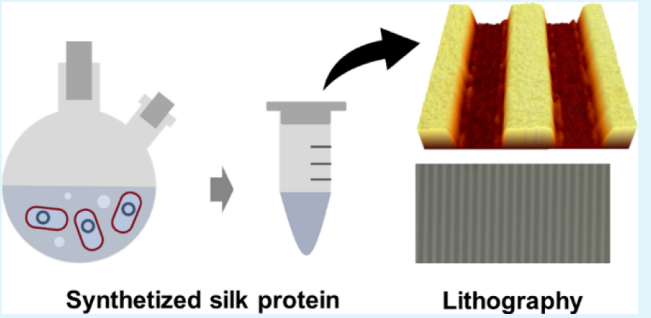
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ABSTRACT: Silk protein is being increasingly introduced as a prospective material for biomedical devices. However, a limited locus to intervene in nature-oriented silk protein makes it challenging to implement on-demand functions to silk. Here, we report how polymorphic transitions are related with molecular structures of artificially synthesized silk protein and design principles to construct a green-lithographic and high-performative protein resist. The repetition number and ratio of two major building blocks in synthesized silk protein are essential to determine the size and content of β-sheet crystallites, and radicals resulting from tyrosine cleavages by the 193 nm laser irradiation induce the β-sheet to α-helix transition. Synthesized silk is designed to exclusively comprise homogeneous building blocks and exhibit high crystallization and tyrosine-richness, thus constituting an excellent basis for developing a high-performance deep-UV photoresist. Additionally, our findings can be conjugated to design an electron-beam resist governed by the different irradiation−protein interaction mechanisms. All synthesis and lithography processes are fully water-based, promising green lithography. Using the engineered silk, a nanopatterned planar color filter showing the reduced angle dependence can be obtained. Our study provides insights into the industrial scale production of silk protein with on-demand functions.



KEYWORDS: *silk protein, synthetic biology, polymorphic transition, lithography, green photoresist*

INTRODUCTION

Silk protein has been showing application potentials toward to the realm of physical technology over realms of biological and biomedical technologies.[1−5] Silk is one of nature-made biomaterials and has been useful for biomedical applications mainly such as wound dressing, artificial nerve guide, and drug delivery due to its biocompatibility, biodegradability, and unique chemical and mechanical properties.[4,6,7] Over the past decade, new approaches for the creation of silk protein-based optical and electrical materials have been developed.[8−10] Using nanofabrication methods, several studies have reported the successful creation of a variety of optical and electrical components including photonic crystals, surface plasmonic crystals, fibers, resistive switching memory devices, and energy harvesting devices.[11−18] Additionally, recent studies have demonstrated that silk protein could be used as an eco-friendly resist for electron-beam and photolithography.[17,18] Particularly, these studies have demonstrated that, due to its unique properties, silk protein could bridge the gap between biological and physical science.

Photolithography is one of the most common patterning methods for the creation of desired shapes at micrometer and nanometer scales using photosensitive materials.[19] Although this method offers several advantages, such as high throughput and resolution, applying this method to soft materials remains an ongoing challenge because an additional photoresist layer is commonly required. Furthermore, it is hard to find etching techniques to transfer the resist soft-mask pattern to the target polymer layer underneath.[20] Additionally, the toxic chemicals in the solvent, developer, and remover solutions used during the etching process limit the biological or chemical functionalization of the target soft material layer. Therefore, alternative lithographic methods such as nanoimprint and dip-pen lithography have been developed to accommodate the processing of soft materials.[20−22] In the last 20 years, lithographic resists with bio-friendly processing requirements

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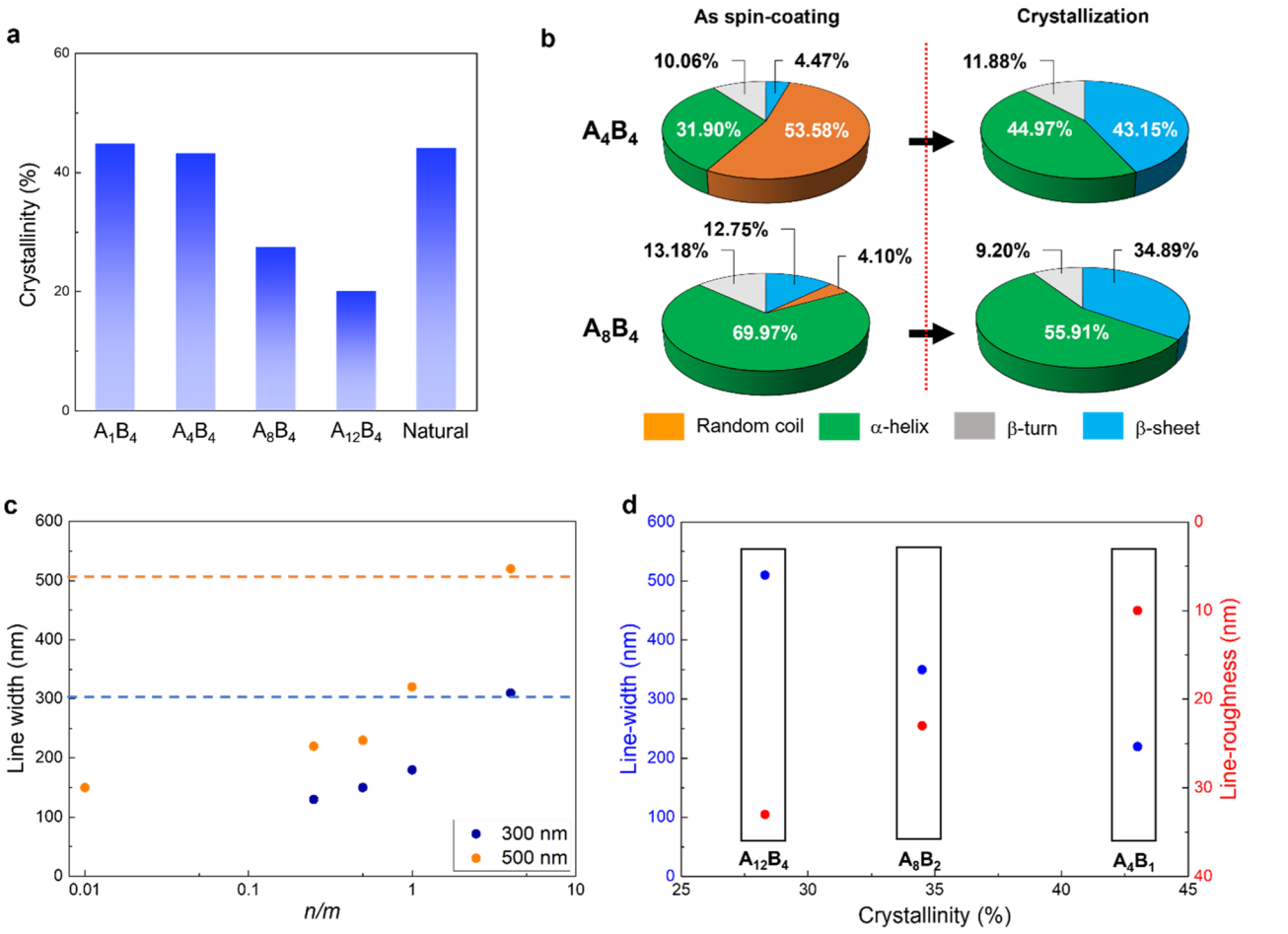


Figure 1. Effects of β-strand and tyrosine-rich domain repetition numbers on polymorphic properties of synthesized silk proteins. (a) Measured crystallinities of the crosslinked silk proteins (A_1B_4, A_4B_4, A_8B_4, $A_{12}B_4$, and natural silk). (b) Pie charts showing the changes in the polymorphic compositions of the A_4B_4 and A_8B_4 silk proteins before and after crystallization. At high hydrophobic domain repetition numbers, secondary structures (mainly α-helixes) were formed upon spin-coating, thus rendering β-sheet induction more difficult. (c) Line widths of the patterns generated by the 193 nm photolithography for various silk types. 300 nm (blue dots) and 500 nm (orange dots) features were targeted. The ratio of the A-domain repetition number (n) and B-domain repletion number (m) was related to the sensitivity of the resists. (d) Measured line width and roughness of the patterned silk resists. Each type of silk protein exhibited different crystallinities.

for patterning biomaterials have also been reported.[23−25] Nevertheless, our previous studies demonstrated that silk protein itself could act as a resist for electron-beam (E-beam) lithography and photolithography, and nanoscale protein patterns could be generated in all-water-based processes in an eco-friendly manner.[17,18] Therefore, the use of silk protein as a photoresist offers a novel means to make the photolithographic method eco-friendly and expand the applicability of the method to biophotonics and bioelectronics.

Nevertheless, an important aspect has been overlooked to use silk protein for biomaterial-based E-beam and photoresist materials and their applications in optical and electrical devices. Unlike inorganic semiconductors and dielectrics typically used in optics and electronics, the synthesis and engineering of biomaterials and the building blocks that comprise them continue to be an important challenge, as the sources of these biomaterials are living creatures.[26] Silk protein is mainly obtained from silkworm cocoons and spiders.[27,28] Although living creatures produce silk proteins with remarkable properties, the molecular building blocks of these proteins are naturally non-homogeneous and cannot be easily changed or manipulated. Synthetic biology can be a key approach to engineer silk protein for the establishment of alternative sources of constant-quality silk on an industrial scale.[29,30] With the insights, in this study, we report the synthesis of silk proteins engineered for green lithography using the 193 nm ArF (argon fluoride) laser and electron beam. Based on computational and experimental analyses, our findings confirmed that controlling the repetition number of β-strand and tyrosine-rich domains of silk proteins affected their crystallinity, the size of β-sheet crystallites, and their photoresponse to the 193 nm deep-UV light, all of which substantially affected the sensitivity and pattern quality of the lithography process. Improving both the quality and photosensitivity of the lithographic pattern, a well-known trade-off parameters in photoresist developments including chemically amplified resists (CARs),[31] was possible using the engineered silk. Our findings could also be applied to design the different type of silk protein suitable for the electron beam lithography. This newly engineered material allowed for eco- and bio-friendly lithographic processing. Moreover, the engineered proteins exhibited good etching resistance, thus allowing for the fabrication of a nanopattern-integrated planar silk color filter with improved angle dependency.

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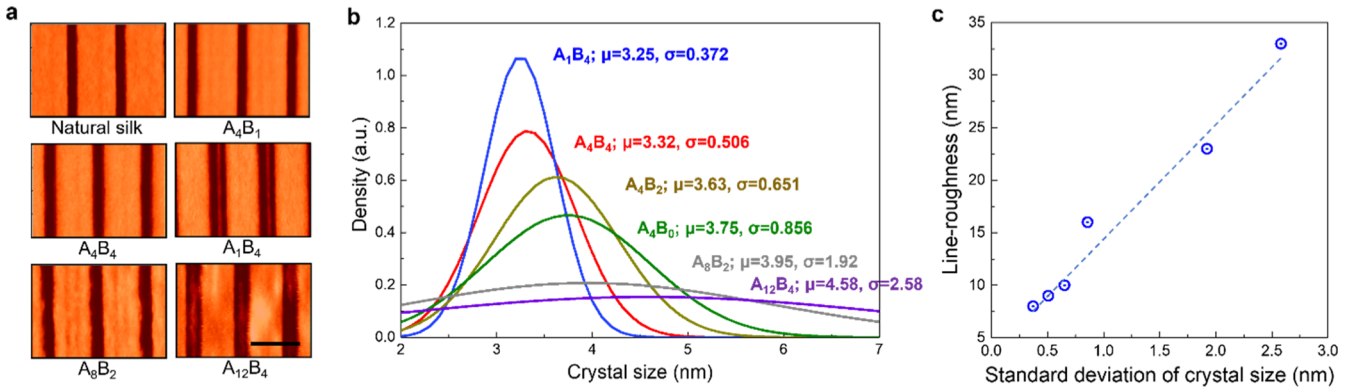


Figure 2. (a) AFM images for the lithographically generated patterns on different silk resists. The scale bar indicates 1.5 μm. (b) Crystal size distribution of the different types of synthesized silks. (c) Line roughness plotted as a function of the standard deviations obtained from (b).

RESULTS AND DISCUSSION

Figure S1 illustrates our proposed silk protein engineering approach for the fabrication of a photoresist for 193 nm lithography. Natural silk protein consists of 11 hydrophilic amorphous regions and highly repetitive crystalline regions (Figure S2).[32,33] These crystalline regions contain three building blocks named "A" (β-strand domain), "B" (tyrosine-rich domain), and "H" (GAAS domain). For the "A" domain (*GAGAGS*), the hydrophobic alanine (*A*) amino acid residues are responsible for the β-sheet stacking, whereas glycine (*G*) facilitates the hydrogen bonding between the β-sheets, resulting in the formation of stiff and strong β-sheet crystals. Tyrosine (*Y*) in the "B" domain is a key amino acid in the self-assembly and crystallization of silk. The $\pi-\pi$ and $\pi-$OH group interactions of tyrosine side chains exert templating effects on the formation of β-sheets.[34] Furthermore, tyrosine reacts with 193 nm UV light, thus enabling the use of silk as a photoresist.[35] Using synthetic biology, silk photoresists with various repetition ratios of "A"−"B" domains (HA_mB_n) were prepared (Figure S2). Controlling the β-sheet size and content and its related polymorphic transitions can be expected. The photoresist was spin-coated on an arbitrary substrate and crystallized using the slow water-vapor process to induce β-sheet formation. Using a photomask, shining the 193 nm laser light on the silk photoresist modified the exposed area, thus rendering it water-soluble via UV-light-induced de-crystallization. Furthermore, only water was used for the development, clarifying, and eco-friendly nanopatterning of the photoresist, making the proposed approach highly eco-friendly.

Line-roughness, resolution, and sensitivity are the main factors to consider when evaluating photoresist performance.[31,36] Nevertheless, photoresist development often entails trade-offs between these parameters. For example, amplification by acid diffusion in CARs enhances sensitivity but increases line-roughness due to random directional diffusion even if CARs have been categorized as the most advanced resist type and have the capability on sub-10 nm resolutive features. However, the use of silk protein with engineered building blocks as a photoresist offers a promising means to overcome the aforementioned trade-offs.[30] Obtaining monodisperse and smaller silk molecules would improve the line-roughness and resolution of the final product. Additionally, increasing the number of photo-reactive amino acid blocks would enhance photosensitivity. In previous studies, natural silk protein exhibited photoresist capabilities that were on par with those of commercial materials.[17,18,37,38] However, engineered silk proteins can be made to be completely monodisperse, whereas natural silk protein had a high polydispersity index (PDI) of ∼7.4. Even CARs often have a PDI of approximately 1.3,[39−41] indicating that silk protein engineering could substantially enhance the precision of lithographic patterning and its related applications.

Structural organization is another important factor when assessing photoresist performance. Before the exposure, spin-coated silk resists were crystallized via water vapor treatment to make the film water-insoluble.[42,43] Given that the underlying mechanism of the lithography was the uncoupling of hydrogen bonding in β-sheets by the UV light, β-sheet content had a strong influence on the resist performance. To investigate the effect of the "A" and "B" regions on β-sheet formation, synthesized silk proteins with repetition numbers *m* and *n* for the "A" and "B" region were spin-coated and crystallized, after which they were analyzed using Fourier transform infrared (FTIR) spectroscopy. Generally, given that β-sheets are known to consist of hydrogen-bonded β-strands in the "A" region, a higher β-sheet content (higher crystallinity) would be expected in silk protein with a greater *m* to *n* ratio. Interestingly, however, our findings demonstrated that the crystallinity was inversely proportional to the repetition number *m* (Figures 1a and S3). The key to understanding the result was the dominant α-helix (69.97%) and β-sheet (12.75%) secondary structures observed in the as-spin-coated A_8B_4 silk before the crystallization (Figure 1b). For the A_4B_4 silk, random coils (53.58%) were the dominant structure (Figure 1b). After the crystallization process, only one-third of α-helices in the A_8B_4 silk (69.97 to 55.91%) turned into β-sheets (12.75 to 34.89%), whereas a seemingly random proportion of the coils in the A_4B_4 silk (4.47 to 43.15%) turned into β-sheets. Chakraborty et al. reported that the transition from α-helices to the β-sheets occurs via collapsed random coil intermediates, and the potential energy barrier for the conversion of random coils to β-sheets is much lower than that to convert α-helices.[44] Therefore, the synthesized silk with the lower *m* value contains more random coils, which favors the induction of β-sheet crystallization. Moreover, the A_1B_4 silk with only one β-strand exhibited similar crystallinity to that of A_4B_4 silk, suggesting that intermolecular hydrogen bonding was remarkably stronger in the engineered silk resist self-assembly than in the intramolecular hydrogen bonds.

Crystallized silk resist films were placed under the photomask and illuminated using a 193 nm ArF excimer laser. The photomask consisted of 100 nm thick chrome (Cr)

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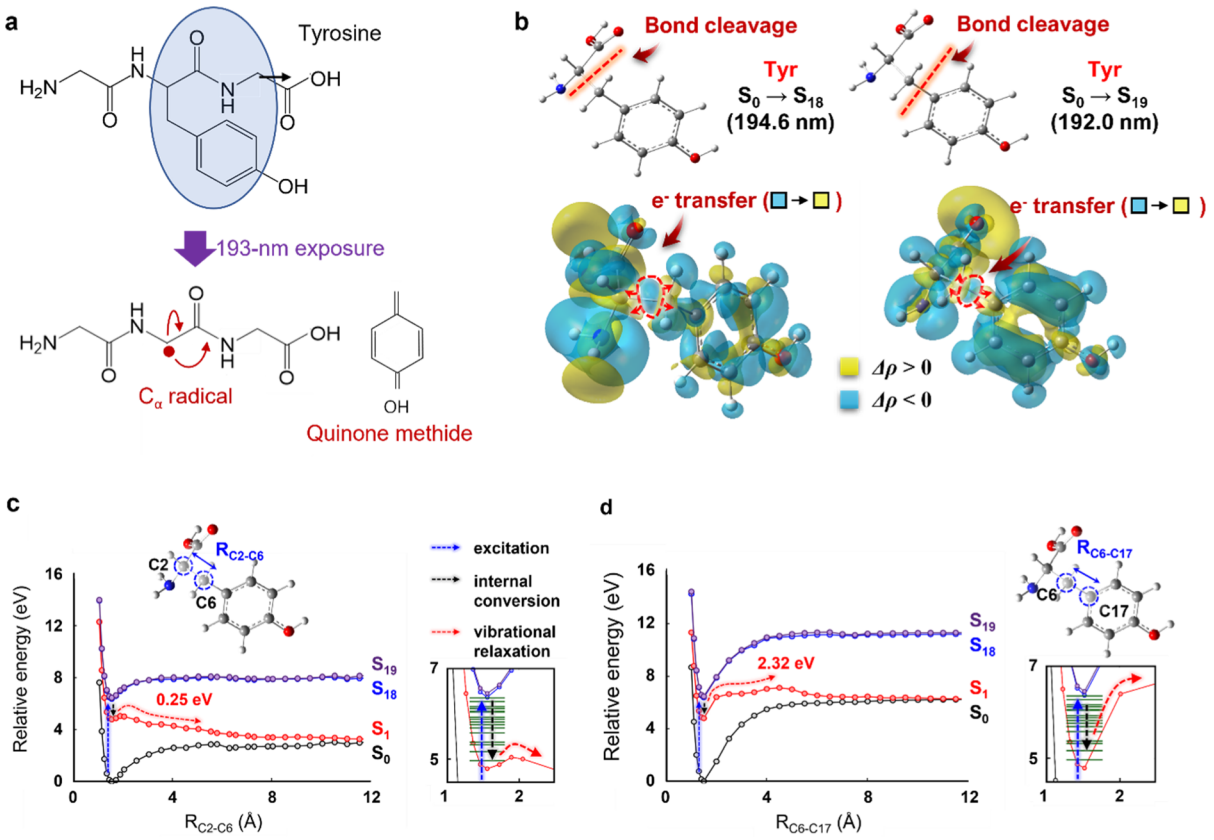


Figure 3. Response of the synthesized silk proteins to the 193 nm deep-UV light. (a) The molecular structure of tyrosine was modeled to demonstrate the radical generation by the light-induced cleavage of chemical bonds. Generation of C_α and QM radicals was expected. (b) Changes in electron densities at the S_0-S_{18} and S_0-S_{19} transitions in tyrosine. The transition energies correspond to the wavelength of the ArF laser (193 nm). The calculations were conducted via the TD-DFT method. From the electron transfer, weakening the strength of the chemical bond between the aromatic ring and main chain. (c) PES profile of the tyrosine group along (a) C2−C6 and C6−C17 covalent bond distance. The arrows indicate the photo-triggered reaction path through excitation, internal conversion, and vibrational relaxation.

lines with various line widths on a fused silica substrate. Upon exposure to 25.44 mJ/cm^2 of radiation, the silk films were developed in water to remove the exposed area. Figure 1c,d shows the measured line widths and line roughness of the synthesized silk proteins. The morphologies of the patterned silk layer were then investigated using atomic force microscopy (AFM) (Figure 2a). Upon measuring the line widths of five types of silk resists (A_4B_0, A_4B_1, A_4B_4, A_1B_4, and natural silk) with similar crystallinity (approximately 44%), the A_1B_4 resist with the highest n/m ratio was found to be the most sensitive and thus accurately reflected the size of the mask pattern (Figure 1c). To investigate the effect of the crystallinity, the line widths and roughness of $A_{12}B_4$, A_8B_2, and A_4B_1 ($m = 3n$ or $4n$) were measured (Figure 1d). Lower crystallinity negatively affected the pattern quality (higher line roughness) and this could be also confirmed by comparing two A_4B_4 silk patterns with different crystallinities (Figure S4a,b).

Another factor to reduce line roughness is the size and size distribution of β-sheet crystallites.[45−47] The size and size distribution of various silk resists were measured using the wide-angle X-ray diffraction (WAXRD) method. Crystal sizes were determined based on the (200) and (210) WAXRD peaks, as they provided information regarding the distances between inter-molecular β-sheets (a) and hydrogen bonding inter-chains (b). For the A_4B_4 and A_4B_0 silks with similar crystallinity, the A_4B_4 resist exhibited broader peaks, which was indicative of a smaller crystal domain and enhanced line-roughness (Figures S4c and S5). Figure 2b and Table S1 show the average sizes and size distributions of β-sheet crystal domains determined by Scherrer's formula.[48] Silk resists with a higher n/m ratio have smaller β-sheet crystal domains and size distributions due to $\pi-\pi$ interactions of the tyrosine side chains. More importantly, this smaller β-sheet crystallite has a narrower size distribution, thus substantially improving line roughness (Figure 2c).

To overcome the trade-off between photosensitivity and pattern quality, the silk resist must be more photo-reactive and possess high crystallinity. The "B" domain contains tyrosine residues, which are highly prone to UV absorption. In turn, this might facilitate the design and fabrication of silk resists with high photosensitivity and good pattern quality. Our findings indicated that the 193 nm UV light could cleave the chemical bonds in tyrosine, resulting in the generation of C_α and quinone methide (QM) radicals that attacked the β-sheets (Figure 3a). Time-dependent density-functional theory (TD-DFT) calculations confirmed that tyrosine was the most responsive amino acid to the 193 nm light (see Figure S6). Furthermore, the resulting electrons moved to the carbon atom from the σ_{C-C} orbital, resulting in QM generation by weakening the bond strength between the phenyl ring and main backbone chain of tyrosine (Figure 3b). The 193 nm UV irradiation excited the electrons in tyrosine to two states: S_{18} (corresponding to 194.6 nm) and S_{19} (192 nm). Internal conversion of excited electrons (to the S_1 state) induced the

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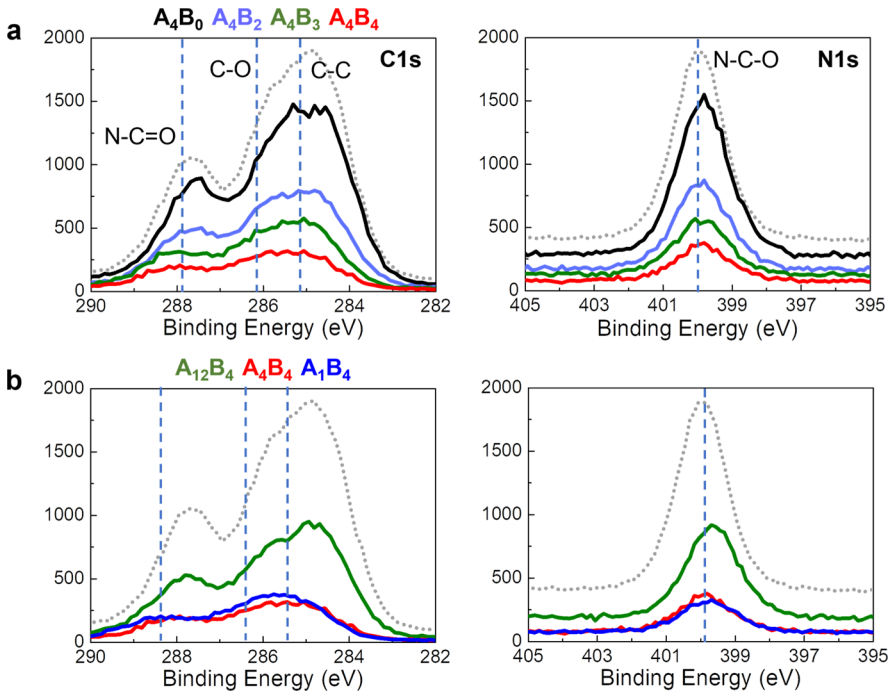


Figure 4. (a,b) XPS spectra measured for different silk types (A_4B_0, A_4B_2, A_4B_3, A_4B_4, $A_{12}B_4$, A_4B_4, and A_1B_4). All samples were irradiated with a 193 nm laser (25.44 mJ/cm^2 dose). The dashed spectral lines correspond to as-prepared silk resist films (before the ArF exposure). The vertical dashed lines indicate the binding energies for the N−C═O, C−O, C−C, and N−C−O bonds.

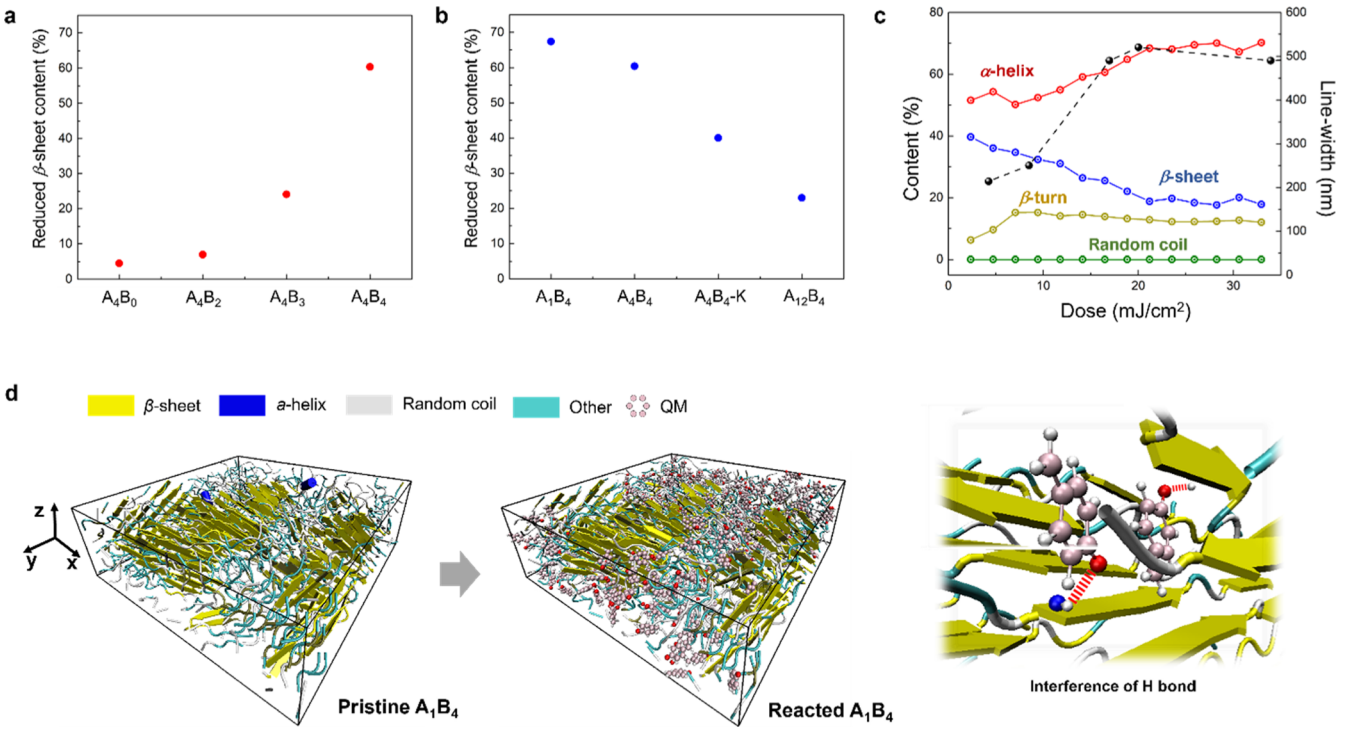


Figure 5. De-crystallization of silk films after the exposure. (a,b) Reduced β-sheet content in different silk resists [A_4B_0, A_4B_2, A_4B_3, and A_4B_4 in (a) and A_1B_4, A_4B_4, A_4B_4−K, and $A_{12}B_4$ in (b)] after 193 nm laser exposure (25.44 mJ/cm^2 dose). "A_4B_4−K" represents A_4B_4 silk to which a lysine residue (a known QM radical quencher) was added. (c) Polymorphic transitions and line widths of the exposed A_1B_4 silk resist as a function of the UV dose. (d,e) MD analysis to characterize the effect of A_1B_4 protein degradation products on the structure of solid films. Configuration of the pristine and reacted A_1B_4 protein models (d). Representative snapshots showing the interaction between the QM radicals and the amides in the β-sheets (left). The irregular A_1B_4 unit domains are not shown for clarity (right).

elongation of the chemical bonds and led to the extraction of radicals (Figure 3c,d). According to our calculations, the 4-hydroxyphenyl radical was preferentially generated over the 4-hydroxybenzyl radical due to its lower extraction energy barrier (0.25 eV, Figure 3c) compared to that of 4-hydroxyphenyl (2.32 eV, Figure 3d) under 193 nm irradiation.

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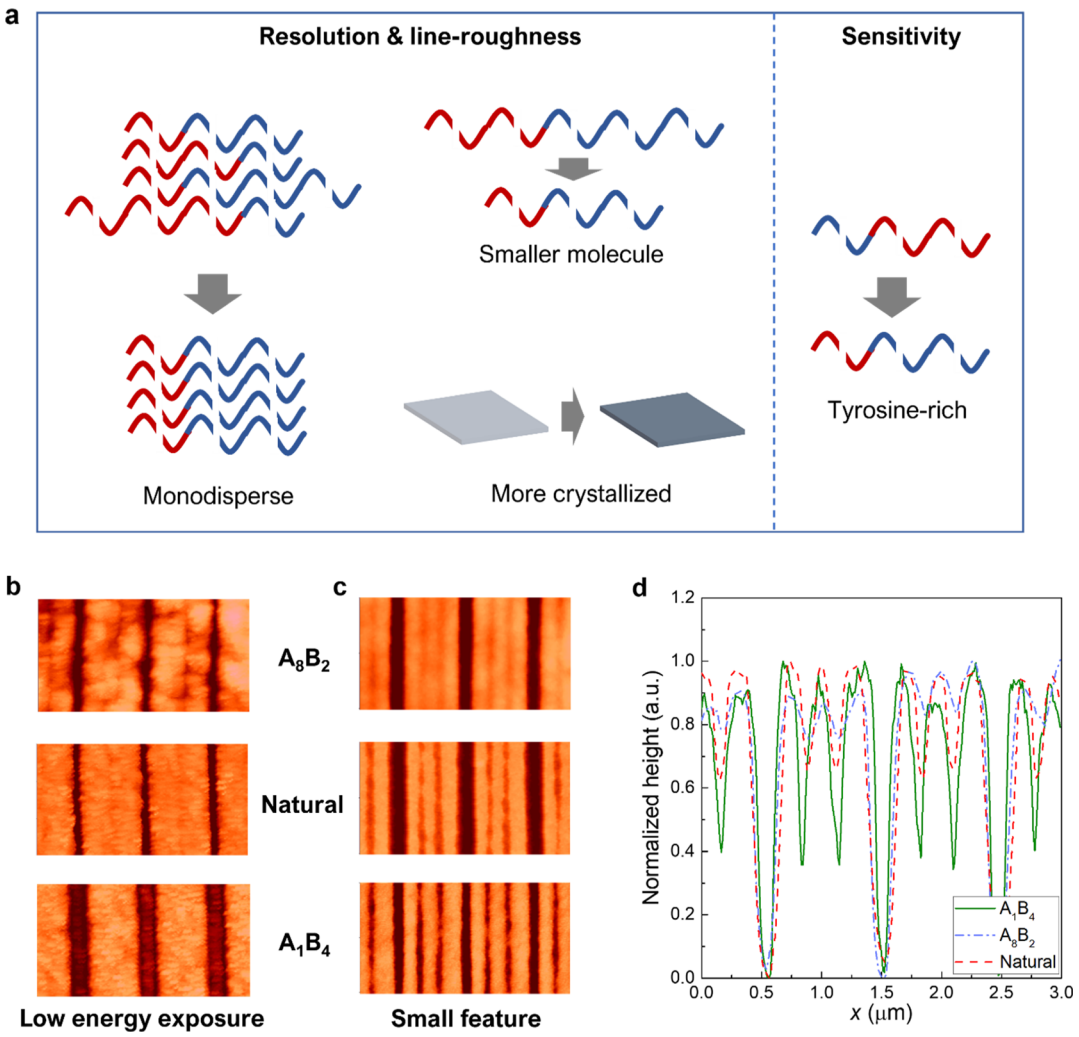


Figure 6. Design parameters to synthesize silk protein for a green lithography resist. (a) Schematic of the requirements for the design of the silk photoresist. Monodisperse silk molecules, smaller crystallites, and film with a higher crystallization degree are necessary to improve resolution and line roughness. Increasing the tyrosine-rich "B" region enhances sensitivity. (b–d) AFM morphologies for the patterned A_1B_4, natural, and A_8B_2 silk resists. With low energy and multi-fringe exposures, the A_1B_4 silk exhibited clear and high-contrast patterns.

As shown in Figure 4a,b, as the relative amount of tyrosine increased (i.e., the n/m ratio increased), radical generation by light-induced chemical bond cleavages also increased. Bond cleavage was further confirmed by the X-ray photoelectron spectroscopy (XPS) measurements (Figure 4a,b). For all silk resist types, the spectral peaks corresponding to the binding energies of N–C=O, C–O, C–C, and N–C–O bonds decreased after UV radiation, thus confirming the cleavage of the chemical bonds. As shown in Figure 4a,b, a more pronounced drop in the spectral peaks was observed as the relative portion of the "B" domain increased (higher n to m ratio), indicating a higher beam sensitivity due to an increase in bond cleavages. Increasing the n to m ratio (either by increasing n while leaving m fixed or decreasing m while fixing n for $A_{12}B_4$, A_4B_4, and A_1B_4) meant that there was a higher relative proportion of tyrosine, which in turn increased radical generation upon UV radiation (resulting in more chemical bond cleavages).

The sensitivity of the silk resists was attributed to the proportion of β-sheets that were de-crystallized due to the tyrosine-generated radicals. We next sought to estimate the polymorphic traits of silk using FTIR spectroscopy. A peak was identified at 1623 cm^{-1} after UV light exposure, which indicated a reduction in the proportion of β-sheets (Figures S7 and 5a,b). For A_4B_0, A_4B_2, A_4B_3, and A_4B_4, the amounts of the de-crystallized β-sheets were proportional to the amounts of the spectral peak drops in the XPS results, meaning that the radicals generated from the chemical bond cleavages affected the β-sheet proportions.[49] The incorporation of lysine (K) to A_4B_4 silk (A_4B_4–K) provided indirect evidence for QM radical generation and its attack on β-sheets. Lysine is a known QM radical quencher due to its stabilizing effect when reacting with the amine group of the latter.[50] Although our XPS measurements indicated that both A_4B_4 and A_4B_4–K generated similar amounts of radicals, the decrease in the number of β-sheets in A_4B_4–K was lower than in A_4B_4 due to the quenching effect of lysine (Figure S8a–c). In turn, this decreased photosensitivity, as demonstrated by the different line widths of the generated patterns (Figure S8d).

Experimental evidence regarding crystallinity, β-sheet crystallite size, and the reduction of β-sheet contents by 193 nm irradiation indicated that the A_1B_4 silk was uniquely suitable as a photoresist material due to its sensitivity and pattern quality. The polymorphic properties of the A_1B_4 silk

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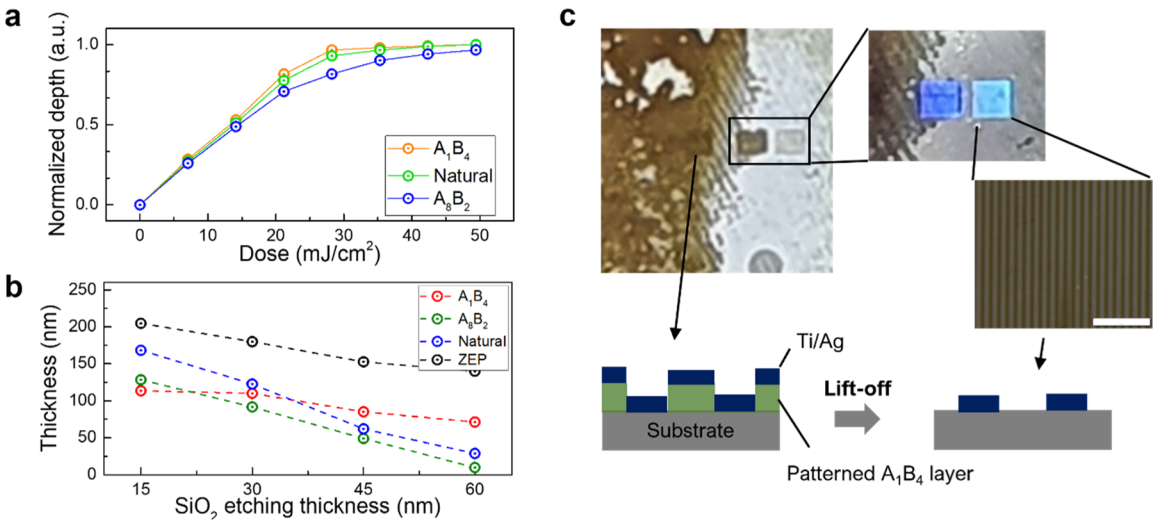


Figure 7. (a) Measured depths on the developed silk resists that were exposed at different energy densities. (b) Thicknesses of silk resists measured after the reactive-ion dry etching process. For comparison, the result for the high etch-resistive commercial resist ZEP was included. (c) Photograph and microscope images and schematic diagram showing the lift-off process applied on the A_1B_4 silk resist. The Ti/Au-coated resist pattern was partially removed to show two patterns (before and after lifted-off) in a single picture for comparison. The scale bar indicates 10 μm.

resist in response to UV dose were analyzed using FTIR (Figure 5c). Our findings demonstrated that the β-sheets progressively turned into α-helixes upon exposure to 193 nm light, whereas β-turn content increased only slightly and remained constant thereafter.

To understand the de-crystallization mechanism, molecular dynamics (MD) simulations were conducted to investigate how the QM radical destabilized and rearranged the β-sheet crystallites. According to our simulations, the QM radicals formed hydrogen bonds with the amide groups in the β-sheet backbone and reduced the interaction energy between the molecular chains that constituted the β-sheets, thus decreasing the proportion of β-sheets.[51,52] Figure 5d shows the structure models of the A_1B_4 silk resist before and after 193 nm light irradiation. The irradiation was modeled by breaking the C2−C6 bonds connecting the side chain and backbone of tyrosine. The separated phenol groups were replaced with QM radicals, whereas the positions of constituent atoms remained unchanged. A remarkable decrease in the β-sheet ratio (~6.7%) and the interaction energy (E_{int}, from −8879.7 to −8653.2 kcal/mol), increase in the number of hydrogen bonds between the QM and the amide in the β-sheet (QuiM/H_n), was observed in the A_1B_4 reaction model (Figures S9 and S10). Based on a representative snapshot, the QM radicals formed hydrogen bonds (QM/H_n) with the amide group in the β-sheet backbone (Figure 5d). These results suggested that the magnitude of E_{int} decreased with the emergence of multiple hydrogen bonds following the dynamics of the QM molecules. Furthermore, MD simulations indicated that the generated QM radicals could reduce the structural stability of the β-sheet crystallite and contribute to the further modification of the secondary structure of the silk resist.

Our study proposes novel engineering approaches for the design of a silk photoresist that combines both pattern quality and sensitivity. Monodisperse and small building blocks that can be highly crystallized provide a better resolution and reduced roughness. Furthermore, the photo-sensitivity to 193 nm laser light can be enhanced by increasing the relative proportion of tyrosine-containing "B" regions (Figure 6a). Therefore, our findings indicated that the A_1B_4 silk protein was more suitable for the construction of a 193 nm photoresist than the other silk resist types tested herein because the A_1B_4 silk protein consisted of small, monodisperse, and relatively tyrosine-rich building blocks. The natural and A_8B_2 silk proteins exhibited narrower and rough patterns when exposed to low energy (~7 mJ/cm^2) through the 500 nm featured mask, whereas the A_1B_4 resist exhibited 500 nm wide and sharp line patterns (Figure 6b). Moreover, the patterning became clearer when diffraction fringes were used. With a fixed power ratio between the strong and weak fringes, the 100 nm feature could be generated only for the A_1B_4 resist (Figure 6c,d). It is worth noting that the engineering parameters can be reformulated to suit the purpose and working principles of the end materials. For example, designing a silk resist for E-beam lithography requires different parameters. In our previous study, we demonstrated that the polypeptides connecting the β-sheets were degraded by high-energy electron irradiance, thus rendering the crystallized silk water-soluble.[17] Therefore, silk protein resists with longer polypeptide chains and lower crystallization were better suited for E-beam lithography, as there was more room for high-energy electrons to break the polypeptide chain. Unlike the 193 nm exposure that induced the chemical reaction of aromatic acids, the A_8B_2 silk exhibited superior lithographic performance compared to A_1B_4 silk in the E-beam lithography (Figure S11).

Contrast curve analyses revealed that the D_{100} (i.e., the dose required for 100% removal) of the A_1B_4 resist was 28 mJ/cm^2, which was substantially lower than that of the natural silk (42 mJ/cm^2) and A_8B_2 resist (over 50 mJ/cm^2), as illustrated in Figure 7a.[53,54] Moreover, the peak power of the illuminating pulsed laser light affected the photoresponsivity of the silk resist (Figure S12), indicating that there was plenty of room to lower the D_{100} value to obtain the designed pattern. One important factor to consider when evaluating photoresist performance is the etching resistance. At 2 sccm O_2 and 40 sccm CF_4 for SiO_2 etching, the A_1B_4 silk resist exhibited an almost 1:1 etching selectivity toward SiO_2. This was better than that for the commercial ZEP520 resist, which is known to have a high etching resistance. In contrast, other silk resists (natural silk and A_8B_2) had much lower resistance (Figure 7b).

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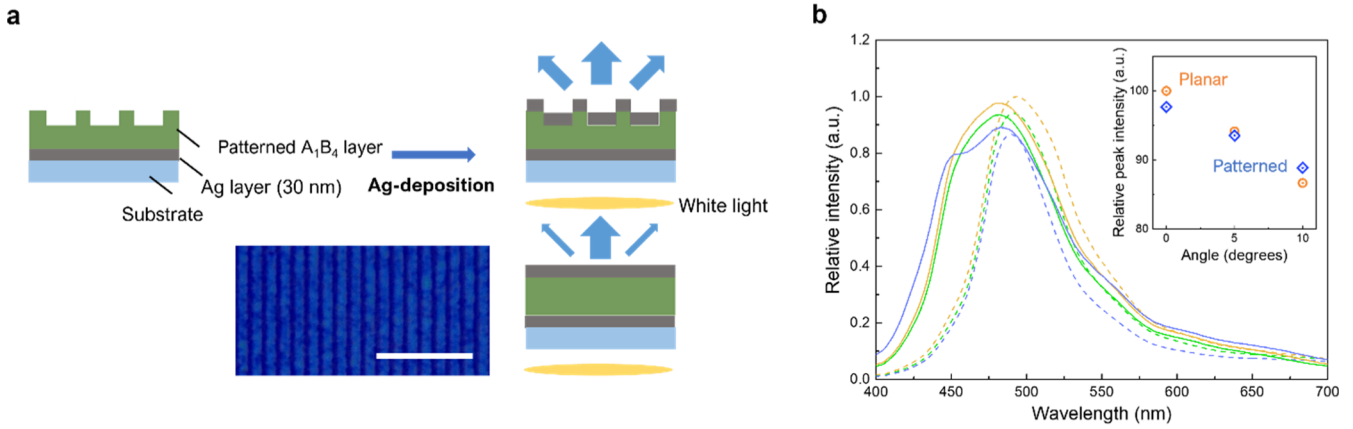


Figure 8. (a) Fabrication process and a microscopy image of the nano FP color filter. The A_1B_4 silk resist was incorporated as the intermediate insulating layer and patterned using 193 nm photolithography. The scale bar indicates 10 μm. Clear grating lines were observed and the blue color indicates that blue light was transmitted through the color filter. (b) Transmission spectra of the nonpatterned (dashed lines) and patterned (solid lines) color filters at different measurement angles of 0° (orange), 5° (green), and 10° (blue). Peak intensities vs the measurement angle are shown in the inset.

The A_1B_4 silk film exhibited a 3.9 GPa Young's modulus, which was higher than that of the natural silk film (2.6 GPa). The higher mechanical strength of the A_1B_4 could potentially explain its etching resistance. Additionally, the Ohnishi parameter (*O*-parameter) of these materials was calculated as follows: *O* = (total number of atoms)/(number of carbon and oxygen atoms).[55,56] Lower *O*-parameter values indicate a higher etching resistance. A_1B_4 and A_8B_2 have *O*-parameter values of 2.28 and 2.34, respectively. Considering that low resistive poly(methyl methacrylate) has an *O*-parameter of 5 whereas high resistive poly(4-hydroxystyrene) has a value of 2.43,[57,58] the low *O*-parameters of silk resists indicate that they are well suited for etching applications with high resistivity while also retaining mechanical strength. However, the increased etch rate at higher O_2 concentrations (1.0 to 1.7 selectivity for 2 to 5 sccm O_2) highlights the fact that etching gases must be carefully selected in etching applications (Figure S13). Additionally, a metallic nanostructure could be generated by the lift-off process. Titanium (Ti) and silver (Ag) layers (5 and 20 nm thick, respectively) were deposited on the patterned A_1B_4 resist, and then, the resist was removed by the lift-off process using lithium bromide aqueous solution (Figure 7c). As a result, a clear Ti/Ag metallic grating on silicon wafer was observed.

A planar metal−insulator−metal resonator based on the Fabry−Pérot (FP) etalon using silk as the intermediate insulating layer could offer versatile sensing and imaging applications as a color filter and absorber with a high quality factor and tunability to water-containing liquids.[59,60] However, although the device structure is simple and can be cost-effectively fabricated, the FP etalon has a poor read-out angle.[61] Integrating nanostructures into the FP resonator affects the FP mode and improves angle dependency tolerance. A 100 nm thick A_1B_4 silk layer was prepared on an Ag deposited glass substrate. The thickness of the Ag layer was 30 nm. Using 193 nm photolithography, a one-dimensional periodic line pattern with a 500 nm width, 45 nm depth, and 1500 nm pitch was fabricated (Figure 8a). White light was illuminated under the sample and the transmitted light was captured with an optical fiber. A blue coloration was observed in the microscopy image, indicating that the transmitted mode through the color filter was located at the blue wavelength as designed. Figure 8b shows the measured transmission spectra at different angles. The nonpatterned color filter presented transmission peaks at 493 nm and a 14% reduction in peak intensity at 10° (inset in Figure 8b). With this pattern, two resonator modes arose. The shorter wavelength mode was blue-shifted, whereas the peak wavelength of the other mode was preserved as the measurement angle increased. Additionally, the reduction of the peak intensity decreased to 9%. Our approach thus allows for the fabrication of the designed nanophotonic structure via mass-production methods using a transparent organic silk protein.

CONCLUSIONS

This study demonstrated the formation of a monodispersed protein block copolymer (A_mB_n) through the genetic engineering of silk protein and its controllable polymorphic change. Polymorphic properties of the engineered silk protein made this material well suited for green lithography resists. Our findings demonstrated how the composition of "A" and "B" blocks in the synthesized silk affected the size, size deviation, and amounts of secondary β-sheet crystallites, and the transition to α-helices by UV-light induced radical generation. Decreasing the *m* value while maintaining the same *n* value increased crystallinity (i.e., increased the proportion of β-sheets) while decreasing crystallite size. In silico simulations and experimental results indicated that the tyrosine residues in the "B" region responded to the 193 nm deep-UV laser pulse and generated radicals, which in turn converted the β-sheet crystallites into α-helix structures (i.e., secondary structures with relatively weak bonds). The polymorphic characteristics of the engineered silk protein endowed the end product with photoresist properties. Particularly, the A_1B_4 silk resist with the smaller and monodisperse β-sheet crystallite and abundant tyrosine amino acids exhibited superior sensitivity, resolution, and line-roughness. Using this silk resist, a narrow 100 nm line was generated, and an almost 1:1 etching selectivity to SiO_2 was obtained under a 30 mJ/cm^2 dose threshold. Selective lift-off process was applied to the silk resist and a clear metallic grating could be obtained. Additionally, the planar color filter could be nano to improve the angle-dependency. Our findings were also applied to the design of the E-beam resist, and we found that A_8B_2 showed the best performance in the E-beam

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case among the different types of silk tested. Taken together, our findings demonstrated that understanding and controlling polymorphic transitions using artificially synthesized silk protein can lead to the development of novel and more effective avenues for the implementation of silk in high-tech applications such as optics and electronics in addition to photolithography.

EXPERIMENTAL SECTION

Silk Protein Expression. Each recombinant plasmid carrying the A_mB_n gene was transformed into *Escherichia coli* BL21 (DE3) (Novagen). Colonies were inoculated into 10 mL of Luria−Bertani (LB) medium with 50 μg/mL kanamycin and grown at 37 °C and 200 rpm overnight. Pre-culture cells were inoculated into 1 L of LB main medium with kanamycin. The cells were then induced with 0.1 mM isopropyl B-D-1-thiogalactopyranoside when the cell density (OD_{600}) reached 0.5, after which they were cultured at 37 °C for 3 h. Cells were harvested by centrifugation and stored at −80 °C for protein purification.

Purification of Synthesized Silk Resists. The cell pellets were re-suspended in buffer A (20 mM Tris−HCl, pH 8.0, 10 mM NaCl). The engineered silk resists were purified using a Ni-NTA column (Qiagen, Chatsworth, CA, USA) according to the manufacturer's instructions. The purified solutions were concentrated at 2% using Amicon Stirred Cells. The purity of the protein in each step was determined by SDS-PAGE using a 12% acrylamide gel stained with Coomassie Brilliant Blue. The purified protein solutions were dialyzed with water using a SnakeSkin (⩽3.5 kDa) dialysis tubing bag (Thermo Fisher Scientific).

Characterization of Silk Resists. The protein solutions were spin-coated on silicon wafers. The thickness of the spin-coated film could be controlled by modifying the spinning speed and the concentration of the silk solution. In this study, a 50 nm thick film could be produced by spin-coating 2% (w/v) silk solution at a maximum speed of 3000 rpm for 40 s. Similarly, a 200 nm thick film was produced by spin-coating 7% (w/v) engineered silk solution at a maximum speed of 1500 rpm for 60 s. The spin-coated samples were soft-baked on a hotplate at 95 °C for 1 min to dry the residual water molecules. The samples were then exposed to water vapor (95 °C at 92% humidity) to induce the crystallization of the silk film.

FTIR analysis of the spin-coated film samples was performed using a Bruker Varian 670 FTIR spectrometer equipped with a Seagull hemi-attenuated total reflection (ATR) instrument. All spectral analyses were conducted using the OPUS 7.0 software. Secondary structures were analyzed by Fourier self-deconvolution (FSD) at the amide I region (1600−1700 cm^{-1}).[62] Second derivatives were acquired with the nine-point smoothing function, and deconvolution was also conducted with the Lorentzian function with a half-bandwidth of 24 cm^{-1} and a reduction factor of 0.24. The FSD spectra were then curve-fitted with the Gaussian function. Curve fitting was then conducted to identify the peak positions of the β-sheet (1610−1630, 1690−1700 cm^{-1}), random coil (1630−1650 cm^{-1}), α-helix (1650−1660 cm^{-1}), and β-turns (1660−1690 cm^{-1}). Secondary structure compositions were estimated by integrating the areas of each fitted curve and then normalizing them to the amide I region area.

The chemical structures of the spin-coated silk resists were determined via XPS analyses (Quantum 2000, ULVAC-PHI) using a monochromatic Al Kα photon source (1486.6 eV) and a 100 μm beam size.

Synchrotron GI-WAXD measurements were performed using the 9A U-SAXS beamline of Pohang Light Source-II. X-rays from an in-vacuum undulator were monochromated using Si(111) double crystals and focused at the detector using a Kirkpatrick−Baez-type mirror. The 2D GI-WAXD patterns were recorded using a 2D CCD (Rayonix MX170-HS). The incidence angles, X-ray wavelength, sample-to-detector distance, and exposure time were set to 0.12°, 1.12 Å (energy = 11.08 keV), 220.242 mm, and 3 s, respectively. To denote the index (*hkl*) and the interplanar distance (*d*) of the specific plane, we assumed that the space group of our vdW SLs was $P6_3/mmc$, which corresponds to that of 2H-phase TMDCs (AA′ stacks).

DFT Simulation. DFT calculations were conducted to examine the photoreactivity of the silk film, and its constituent amino acids were tested, including glycine (Gly), alanine (Ala), serine (Ser), tyrosine (Tyr), and valine (Val). Here, the Gaussian 16 packages[63] were utilized for molecular modeling and electronic structure characterization. Our calculations were based on the B3LYP/6-311+G(d,p)[64,65] density functional theory, which is a widely accepted approach for describing the electronic transition of organic compounds.[66,67] The net charge and spin multiplicity of the amino molecules were neutral and singlet, respectively.

In the present study, photo-response simulations were conducted in two steps: (1) the photosensitive amino acids among the molecular components of the silk film were sorted out; (2) the photo-triggered chemical reaction path of the corresponding monomer was identified.

Regarding the photosensitivity of the silk protein components, the ultraviolet−visible (UV−vis) absorption spectra of the amino molecules were computed using time-dependent DFT (TDDFT),[68] after which fully optimized molecular structures were employed. The most responsive monomer and the associated reaction coordinate upon light irradiation were determined based on the oscillator strength and the electron density difference.

The next step is to predict the potential energy surface (PES) for the photo-response of the target amino acid. During the ground state PES scan, the amino structure was relaxed with the constrained reaction coordinate. An excited state energy surface was determined via TDDFT calculations for every scanning sample in the ground state, which mimics the vertical excitation of the electrons. The activation energy is the key parameter to determine the major electronic transition and the subsequent chemical reaction path for the target monomer under the ArF light source.

Laser Exposure for Lithography. The photoresists were exposed using a 193 nm deep-UV ArF excimer laser (ExciStar 500, Coherent) via a photomask. The laser system had a pulse duration of 7 ± 2 ns.

Fabrication and Characterization of the FP Resonator. FP color filters were fabricated on a quartz substrate. An electron beam evaporator (SEE-7, ULTECH) was used to deposit the bottom Ag layer (30 nm thickness), followed by spin-coating of the intermediate A_1B_4 silk layer. After crystallization, the top Ag layer (30 nm thickness) was deposited on the intermediate layer using the same evaporation method. For transmission measurements, a halogen white light was illuminated directly onto one surface of the resonator, and the transmitted light was collected from both the planar and patterned regions using an optical fiber placed near the opposite side of the resonator. The optical fiber was coupled to an optical spectrum analyzer to conduct the measurements (USB-2000+VIS-NIR-ES, Ocean Optics).

Morphology Characterization of the Silk Resists. Clear lines and the transmitted color of the lithographically generated pattern were observed using an optical microscope (BX53M, Olympus). The objective lens used in this study had a numerical aperture of 1.3 and a 100× magnification. The surface morphology and line-roughness of the silk patterns were also measured using AFM (XE-100, Park Systems). The two-dimensional step heights of the patterned and non-patterned sample surfaces were measured using a surface profilometer (Tencor P-10 Stylus Profiler, KLA Instruments) with a 5 nm to 130 μm measuring range. The thickness of the etched samples was measured using an ellipsometer (Film Sense FS-1). The ellipsometer could accommodate samples up to 200 mm in diameter and 23 mm in thickness and could measure film thicknesses ranging from 0 to 1000 nm with a <0.001 nm precision at a 1 s acquisition period.

MD Simulation. All MD simulations were conducted with the Materials Studio software (Dassault Systems, 2020) using the COMPASS III force field.[69] As illustrated in Figure S2, the fully extended A_1B_4 chain consisting of a GAGAGSGAASGAGAGSGA-GAGYGAGVGAGYGAGYGAGAGAGY sequence (one-letter amino acid code) was prepared. The single protein unit was composed of 16

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A_1B_4 chains and was modeled with the following procedure: A_1B_4 chains were evenly arranged in a 4 by 4 array at a 5 Å distance. The first half of the sequence (GAGAGSGAASGAGAGSGAGAG) was placed in an antiparallel manner to construct the β-sheet, whereas the Tyr-rich sequence (YGAGVGAGYGAGYGAGAGAGY) followed an irregular pattern. The positions of the atoms were optimized using the conjugate gradient method with a cutoff of 10 kcal/mol and then subjected to an *NVT* (isothermal−isochoric) ensemble at 300 K for 10 ns at 0.5 fs time steps. Four A_1B_4 protein units were randomly located in a periodic cubic box with a density of 0.2 g/cm^3 and then subjected to an *NPT* (isothermal−isobaric) ensemble at 300 K for 3.5 ns, which resulted in gradual densification of the entire system. The converged density of the system after equilibration in the *NPT* ensemble was 1.243 g/cm^3. Undesired damages induced by sudden contact with adjacent protein units were restored by replacing the β-sheets of the protein unit with the pristine ones after the process was completed. Subsequently, additional equilibration in the *NVT* ensemble was performed at 300 K for 300 ps, followed by the *NPT* ensemble at 300 K and 1 atm for 200 ps. The completed solid film model had a length of 9.3 × 9.3 × 3.0 (nm^3) in each direction and a density of 1.283 g/cm^3. Finally, the thermodynamic equilibrium of the model as well as the stability of its microstructure was confirmed by running the *NVT* ensemble at 300 K for 3.5 ns. The morphological changes in the A_1B_4 protein induced by ArF laser irradiation were attributed to the breakage of the C−C bonds connecting the side chain and the backbone of tyrosine. The phenol groups separated from the A_1B_4 molecules were replaced with QM, while the position of the constituent atoms remained unchanged. Afterward, an *NVE* (microcanonical) ensemble was performed for 5 ns with an additional energy of 432.64 eV to account for the exothermic behavior of tyrosine during the photoreaction. The secondary structure of the A_1B_4 protein was analyzed based on the STRIDE algorithm using the open-source visual MD (VMD) software.[70] The structural analysis results for the reacted A_1B_4 was obtained by averaging the last 2 ns of the *NVE* ensemble. We also measured the interaction energy (E_{int}, defined as the non-bonded energy that each A_1B_4 molecule forms with another A_1B_4 molecule in the nearest region) between molecular chains constituting the β-sheet domain. The number of hydrogen bonds between the QuiM and the β-sheet backbone (QuiM/H_n) was counted using a distance cutoff of 3.5 Å and a 90° angle.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acsami.2c17843.

Schematic for engineering of silk protein molecules, mechanism for synthetized silk protein, polymorphic properties of the engineered silk proteins, effects of crystallinity of line width and roughness, XRD pattern for the silk resist film, simulation results calculated by the density function theory on silk protein, data to show the polymorphic transitions for various types of silk resists, influence of the lysine quencher on QM radical generation, simulated molecular structure of the A_1B_4 silk, MD study to understand the effect of the degradation products of the A_1B_4 protein on the structure of solid films, electron-beam lithography results, plot to show the effect of peak power on lithographic performance, etch rate of the silk resist, and crystallite size values of silk resists (PDF)

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Author Contributions

S.-C.C., G.K., and S.K. designed the experiments. J.-S.P., J.K., and J.K. performed engineering the building blocks of silk protein and expression of synthetic silk protein and fabricated the silk film. J.K. and J.K. analyzed the FTIR data and XPS data. J.-S.P. performed the crystal size and distribution of silk film. M.K. performed the TD-DFT simulation. H.K. and J.C. performed the MD simulation. R.K.J. and J.C. made the lithography setup and performed the lithography processes. R.K.J. fabricated the color filter. R.K.J., J.C., N.G., and J.-Y.P. performed the optical and AFM investigation for the generated patterns and devices. T.-Y.L. and H.J. prepared the photomask and performed the SiO_2 etching process. S.-C.C., G.K., and S.K. wrote the manuscript.

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Notes

The authors declare no competing financial interest.

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